



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Diseases of Myelin

Principles and Practices of Osteopathic Medicine I

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

- 1. With a background of demyelination pathogenesis, discuss and describe the pathological, morphological, histological, and clinicopathological features of **multiple sclerosis**.
- 2. Discuss and describe the pathogenesis, morphological, histological, and clinicopathological features of **Guillian-Barre syndrome**.
- 3. Explain and compare the pathogenesis, morphological, histological, and clinicopathological features of selected **leukodystrophies including metachromatic leukodystrophy, Krabbe's disease, and adrenoleukodystrophy**.
- 4. Discuss the pathogenesis, morphological, histological, and clinicopathological features of **progressive multifocal leukencephalopathy**.

SELECTED TOPICS

Inflammatory Demyelinating Diseases

- CNS: Multiple Sclerosis
- PNS: Guillain-Barre Syndrome

Leukodystrophies (CNS & PNS Dysmyelinating Disorders)

- Metachromatic Leukodystrophy (Sulfatide lipidosis)
- Krabbe's Disease (Globoid Cell Leukodystrophy)
- Adrenoleukodystrophy

Progressive Multifocal Leukoencephalopathy (PML) (Demyelination with an infectious etiology)

Basics of Demyelination

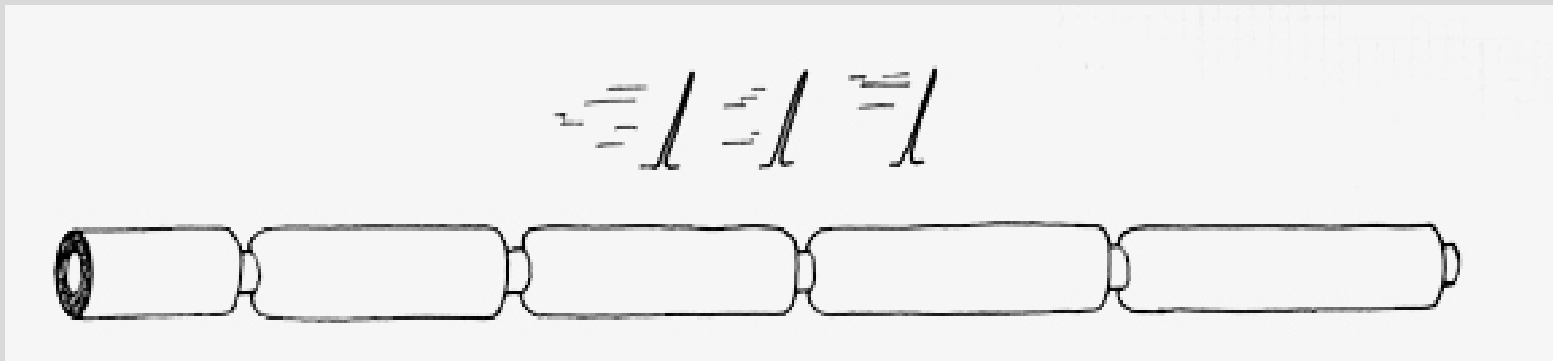
1. Loss of myelin (demyelination)
2. Relative preservation of axons

- **Preservation of axons** key to help distinguish demyelinating process from more nonspecific destructive process
- Both an infarct and a demyelinating lesion have loss of myelin and show foamy macrophages phagocytizing the debris.
- In an infarct, however, the axons will also be destroyed.

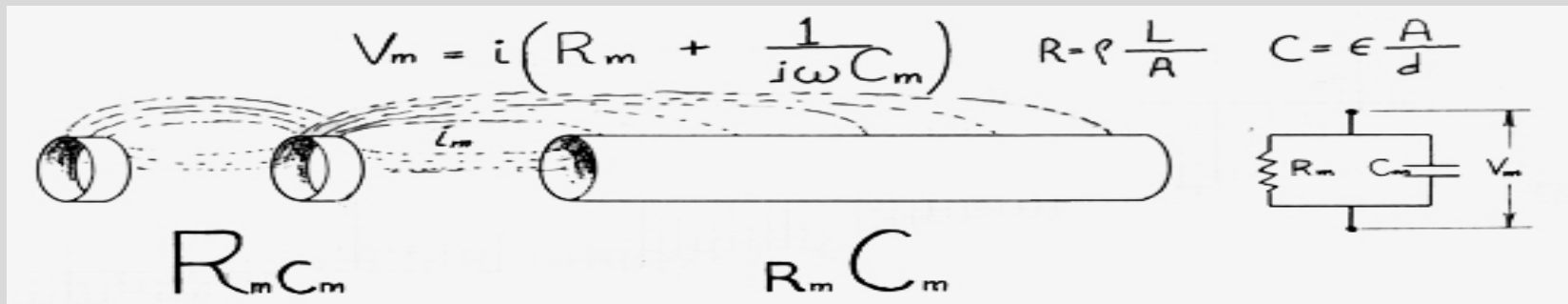
Basics of Demyelination

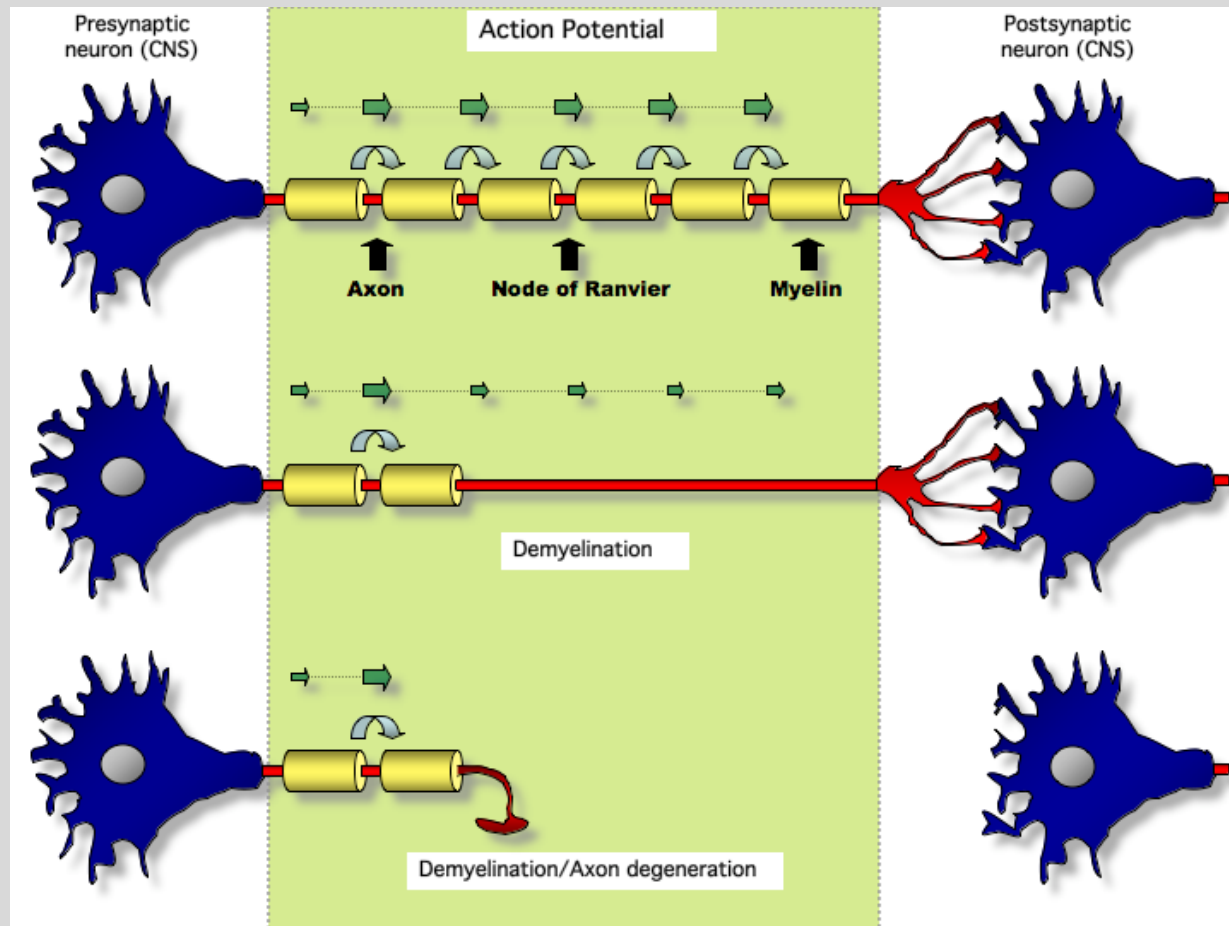
Effects of demyelination

Slowing of conduction velocity due to loss of saltatory conduction.



Conduction block due to impedance mismatch between myelinated and unmyelinated portions of the axon.





The **velocity** with which an action potential travels down the axon depends on (a) **whether the fiber is myelinated** and (b) the **diameter** of the fiber. Conduction by local current flow does occur in unmyelinated fibers. However, a faster method of propagation, saltatory conduction, takes place in myelinated fibers.

Case Question

- A 33 year old woman complains of urinary incontinence and **blurry vision** for the past 3 months. A fundoscopic exam shows no abnormalities. Two months later she develops double vision and numbness in the fingers of her left hand. MRI shows **scattered plaques** in the patient's brain and spinal cord. Over the next several months, some of these plaques diminish in size, while others appear in new locations. These plaques would most likely show selective loss of which of the following proteins?
 - A. beta-amyloid
 - B. glial fibrillary acidic protein
 - C. synaptophysin
 - D. myelin
 - E. alpha-synuclein

Multiple Sclerosis

- (Chronic) **Autoimmune demyelinating** disorder with distinct episodes of neurologic deficits separated in time and attributable to patchy white matter lesions separated in space
- Most common demyelinating disorder
- Affects any age though onset in childhood or after 50 years of age is rare
- Women are affected twice as often as men
- Clinical course - relapsing and remitting episodes of variable duration (weeks, months, years) marked by neurologic defects, followed by gradual and partial recovery of neurologic function

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch.28

Multiple Sclerosis - Pathogenesis

- **Autoimmune response against components of myelin sheath**
- Involves genetic and environmental factors
- Incidence of MS 15-fold higher if present in first-degree relative
- Strong association with a DR haplotype of the major histocompatibility complex
- Other associations noted with IL-2 and IL-7 receptor genes, and genes encoding proteins involved in immune response, including cytokines and their receptors, and cytoplasmic signaling molecules
- Environmental factors - **geographic variation in prevalence of MS, higher number of cases diagnosed away from the equator;** latitude dependence may be related to low level of vitamin D (an immune system modulator) in people who are not exposed to sunlight during winter

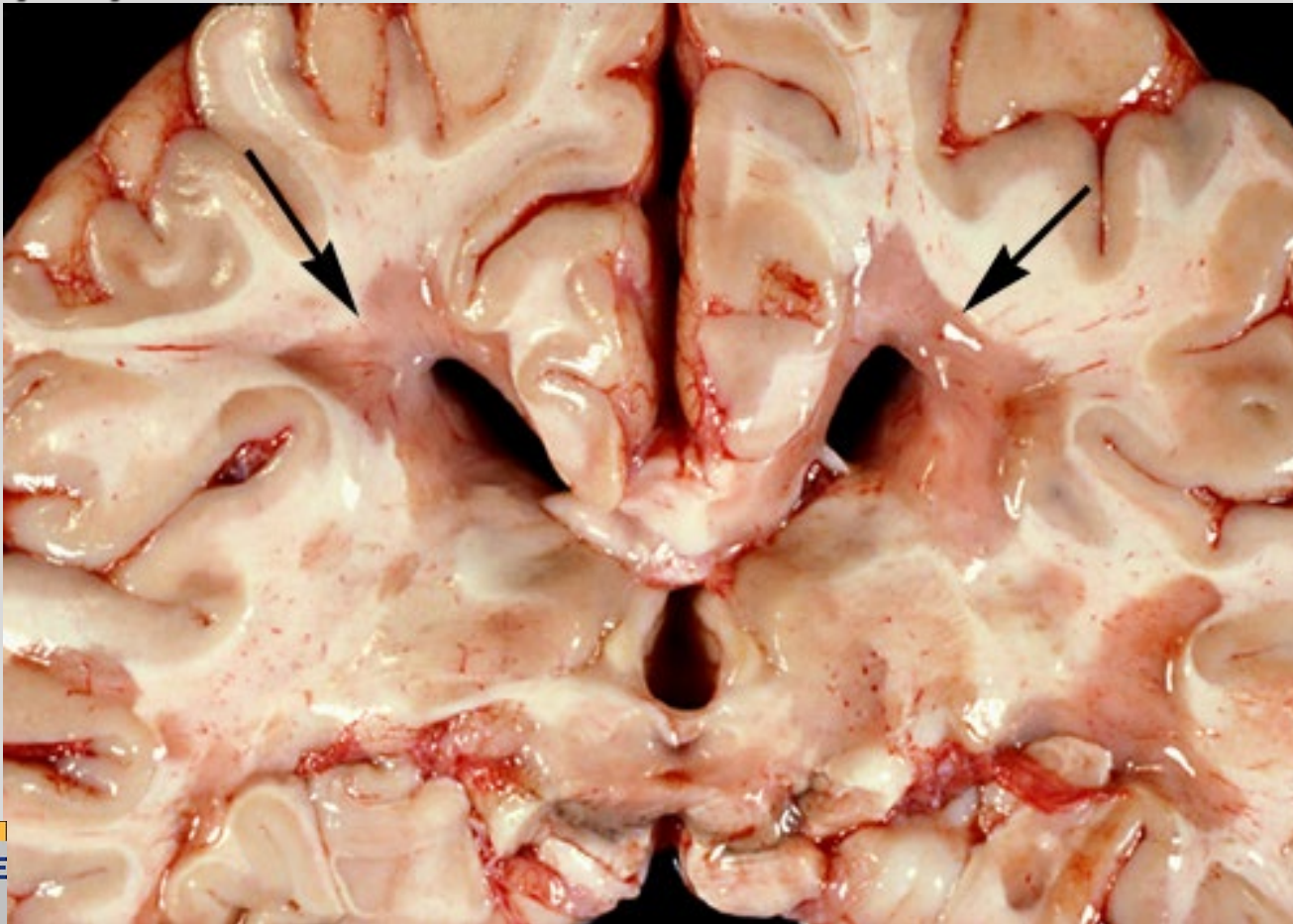
Multiple Sclerosis - Pathogenesis

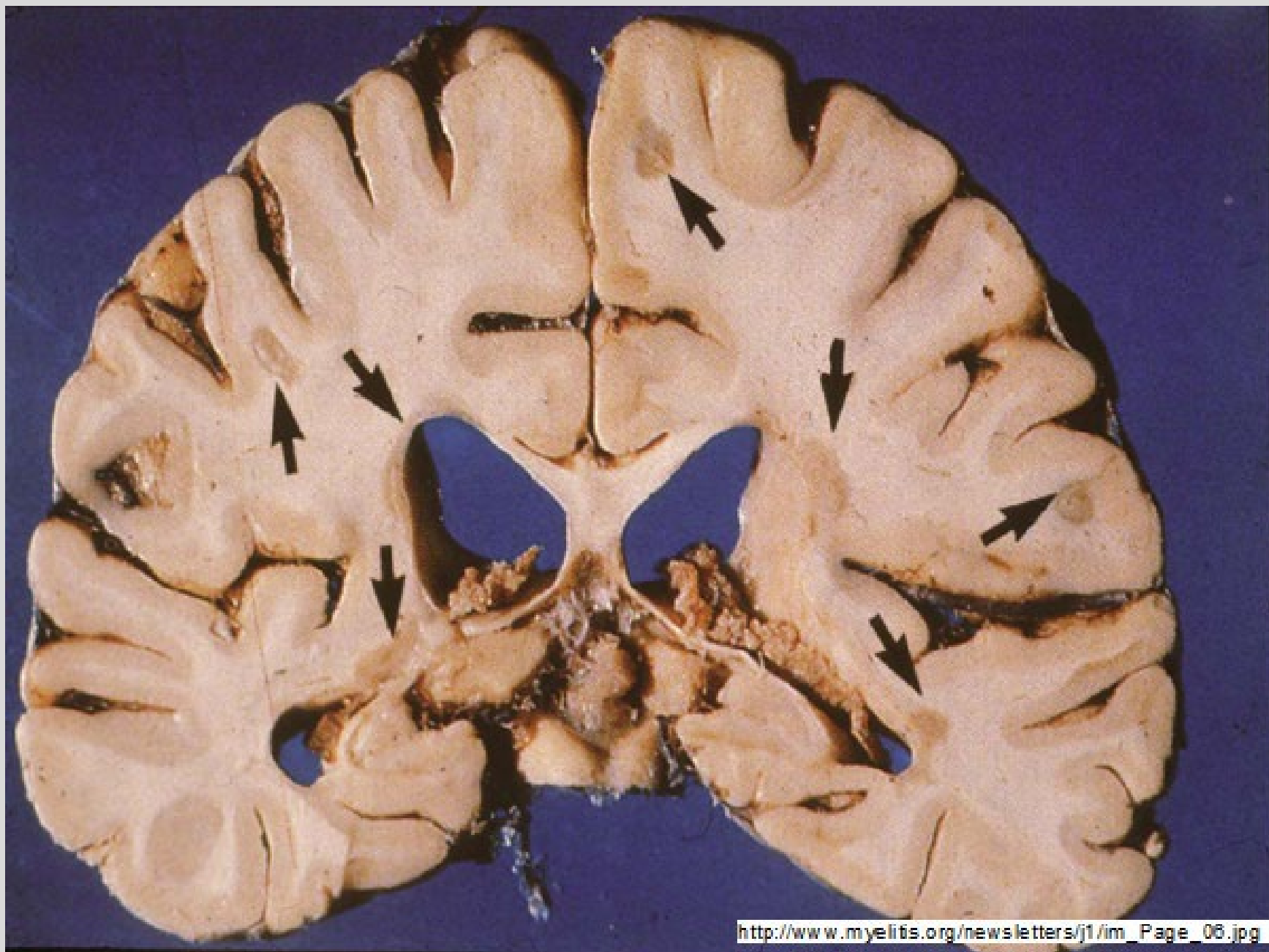
- Available evidence: MS initiated by Th1 and Th17 T cells that react against myelin antigens and secrete cytokines
- Th1 cells secrete IFN- γ , which activates macrophages
- Th17 cells promote recruitment of leukocytes
- Demyelination caused by these activated leukocytes and their injurious products
- Infiltrate in plaques and surrounding regions of the brain consists of T cells (mainly CD4+, some CD8+) and macrophages
- Therapies being developed that modulate or inhibit T-cell responses and block the recruitment of T cells to the brain
- Contribution of humoral immunity suspected, based on oligoclonal bands of immunoglobulin in CSF
- MS appears to be an **autoimmune disease** triggered by an **environmental factor** in a **genetically susceptible** individual.

MS – Gross findings

- **White matter disease**
- In fresh state, lesions are firmer than surrounding white matter (“sclerosis”) - well circumscribed, somewhat depressed, glassy, gray-tan, irregularly shaped plaques
- **Area of demyelination often has sharply defined borders**
- Size of lesions varies from small foci (microscopic) to large confluent plaques
- Plaques commonly occur adjacent to the lateral ventricles and also often in the corpus callosum, optic nerves and chiasm, brainstem, ascending and descending fiber tracts, cerebellum, and spinal cord.
- Plaques can extend into gray matter because myelinated fibers are present there as well.

Periventricular, symmetric **MS PLAQUES** by the lateral ventricles (arrows): demarcated grey/ grey-pink lesions in unfixed brain tissue; firmer than the adjacent normal white matter (*sclerosis*). Plaques are often multiple and frequently symmetrical





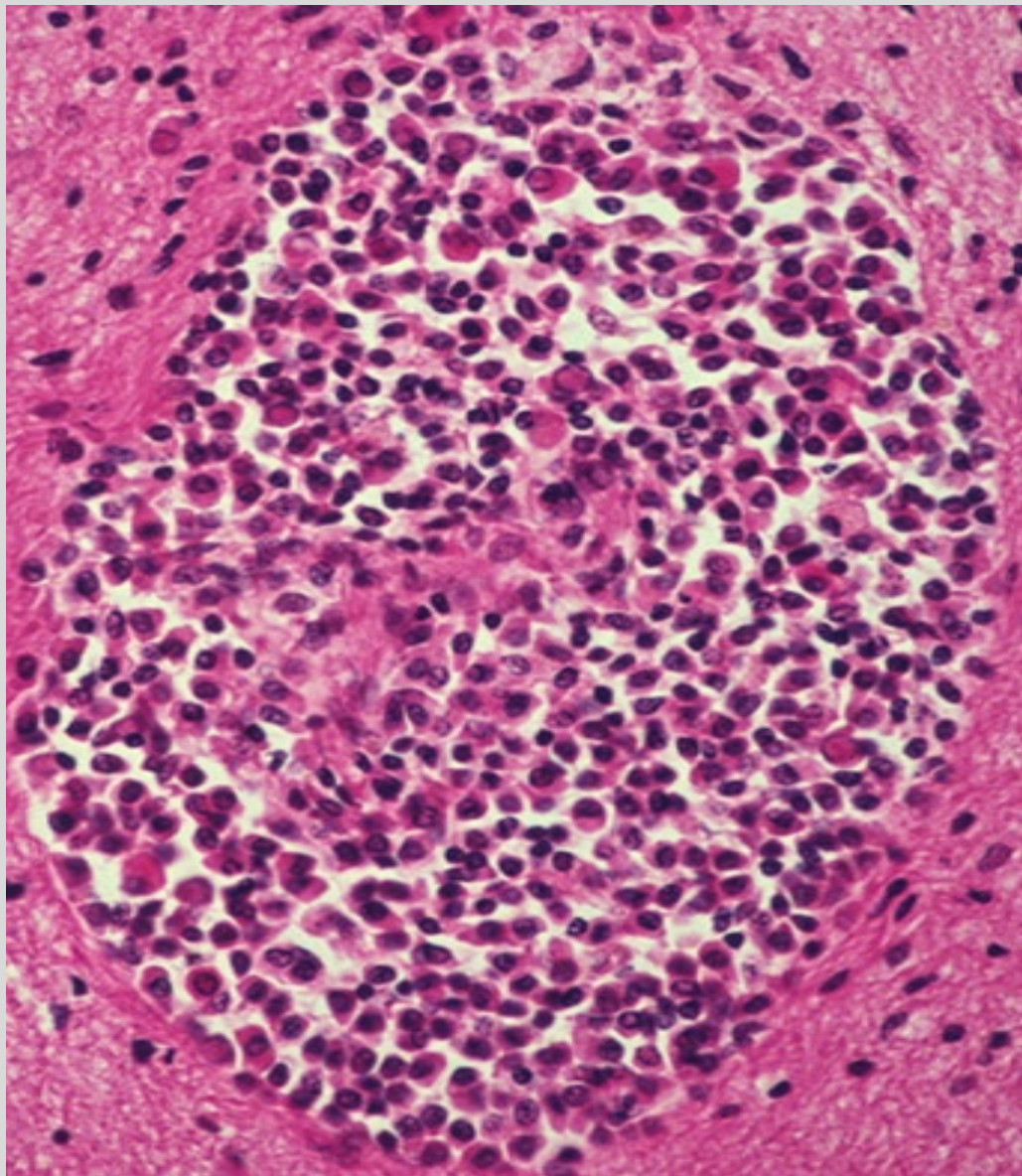
MS Microscopic Pathology

- Active plaque - **ongoing myelin breakdown associated with abundant foamy macrophages**; lymphocytes also present, mostly as perivascular cuffs, especially at outer edge of the lesion.
- Active lesions often centered on small veins; myelin is usually completely absent; axons are relatively preserved
- In time, astrocytes undergo reactive changes.
- As lesions become quiescent, the inflammatory cells slowly disappear.
- Within **inactive plaques**, there is no macrophage-rich infiltrate, little to no myelin is found, and there is a reduction in the number of oligodendrocyte nuclei; instead, **reactive gliosis** is prominent
- Note – Active – “acute” / Inactive – “chronic”

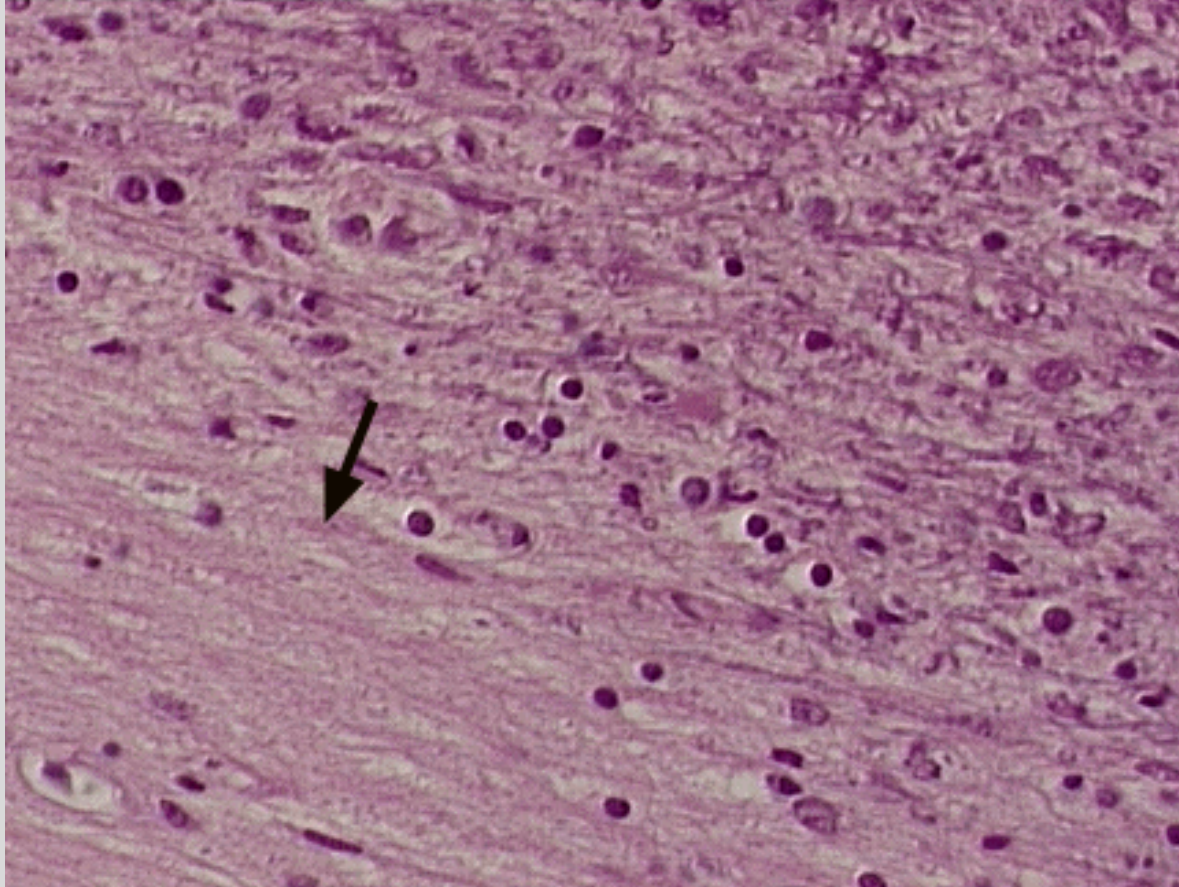
MS Histology

- MS lesions vary depending on their age.
- *Early (active/acute) lesions :*
- perivascular and parenchymal infiltration by **inflammatory mononuclear cells**
- **myelin breakdown** and phagocytosis by macrophages
- astrogliosis not yet prominent and axons relatively preserved.

As lesion progresses, fewer inflammatory cells and more astrogliosis



http://missinglink.ucsf.edu/lm/ids_104_Demyelination/Didactic/MSMicroPath.htm



High power view of demyelinated plaque showing decrease in oligodendroglia and increase in the larger paler astrocytic nuclei and the fibrillar background due to gliosis.

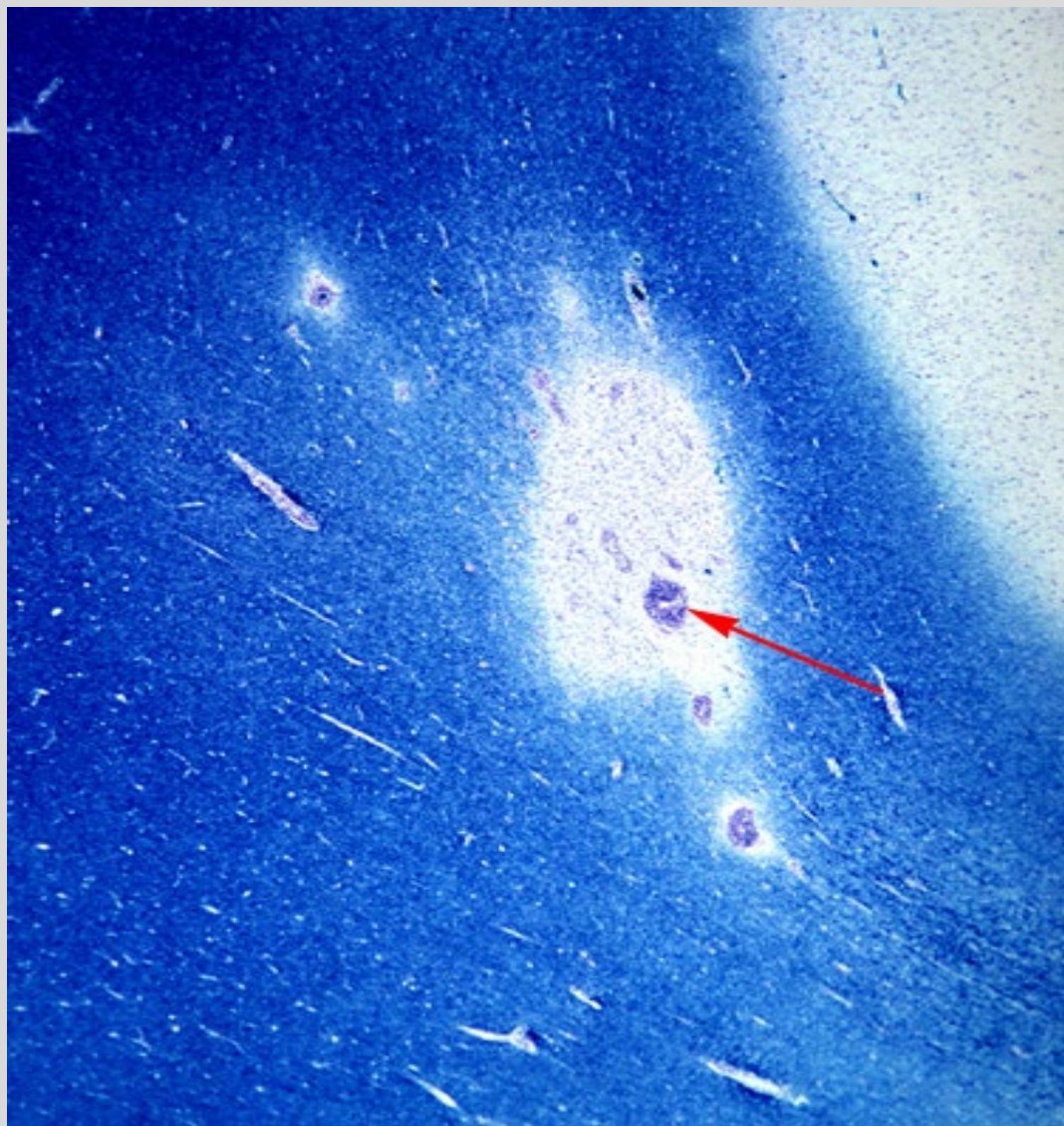
- This plaque is near a ventricle, a common place for an MS plaque.

(Description By: Margaret Grunnet, M.D.)

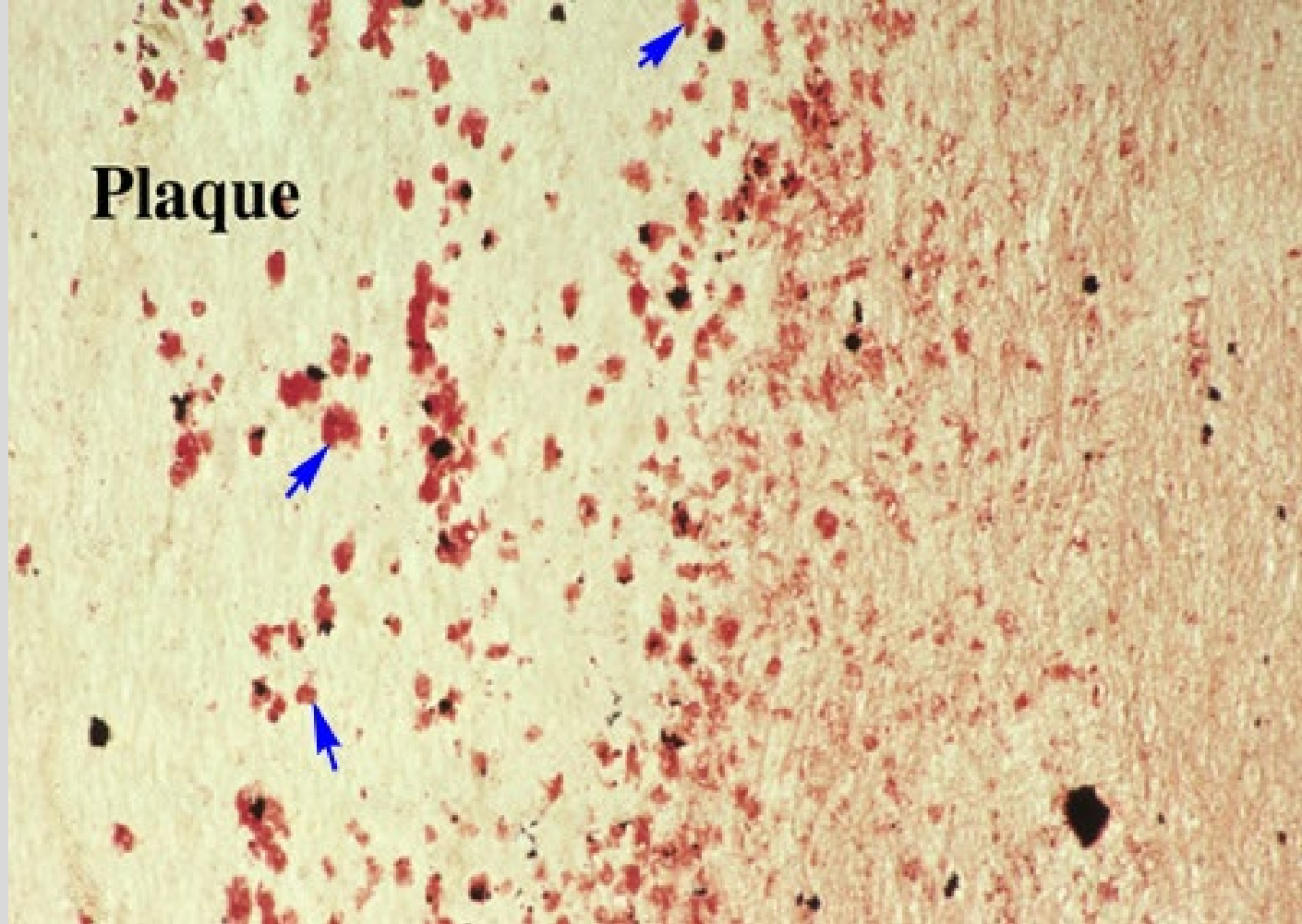
(Image Contrib. by: Margaret Grunnet, M.D. UCHC)

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Early lesion centered around a small vein (arrow)
which is surrounded by inflammatory cells. (luxol fast
blue/PAS)



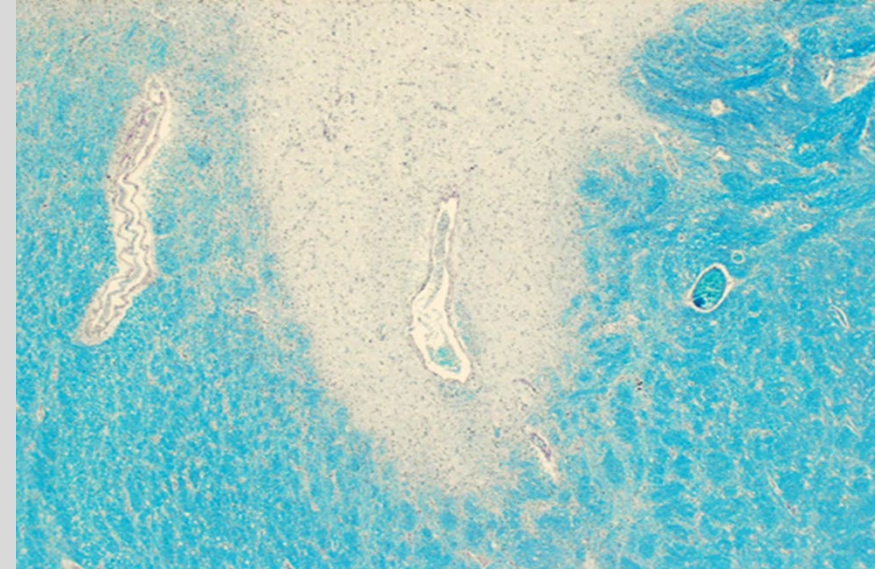
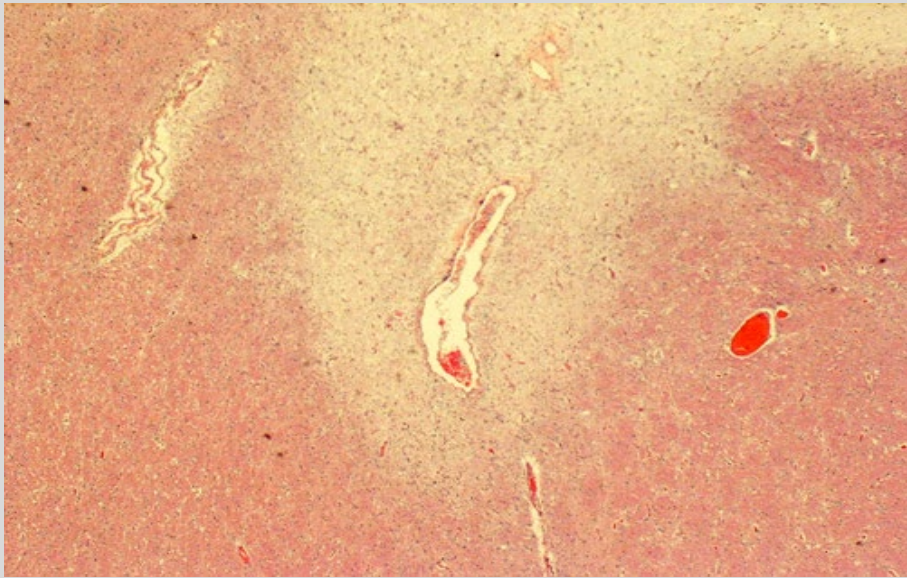
Oil Red O-stain: edge of lesion (plaque) on left with more normal white matter on right. Lipid-laden macrophages in lesion (arrows) and at interface.

MULTIPLE SCLEROSIS OLDER PLAQUES

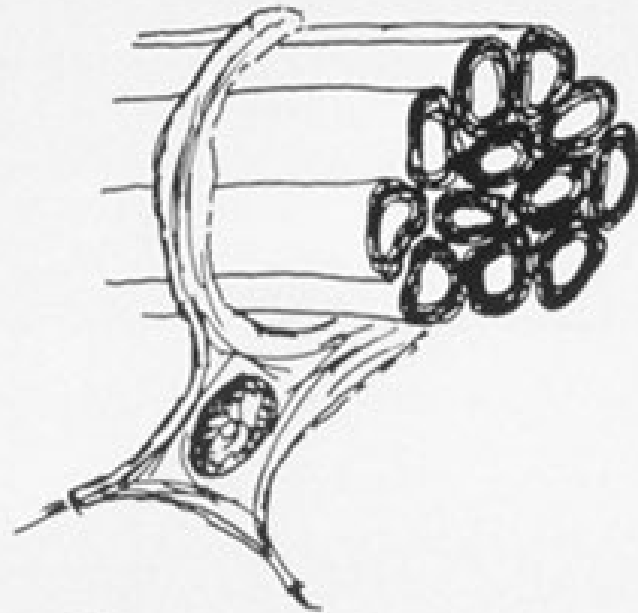
Chronic lesions :

- Few mononuclear cells
- Almost complete demyelination
- Severe astrogliosis
- May be oligodendrocyte loss and some secondary axonal loss in **advanced** cases

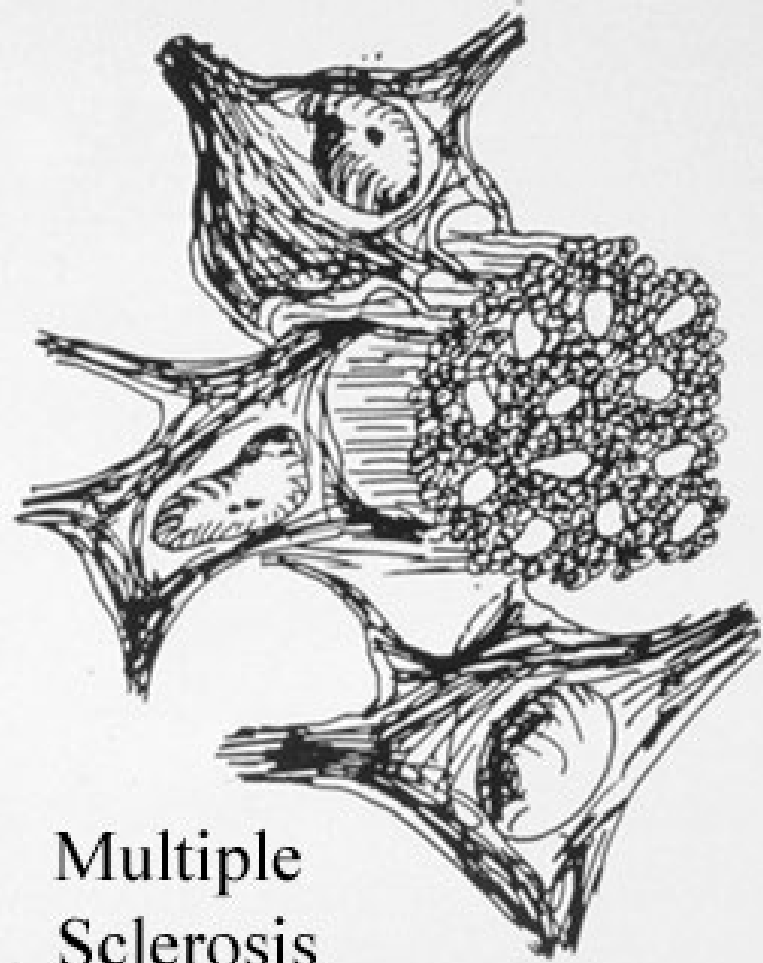
- **Periventricular lesion centered around vein.**
- **Older lesion: very little inflammation around the vein.**
- **H and E loss of myelin: lighter pink than surrounding normal white matter (left photo)**



http://missinglink.ucsf.edu/lm/ids_104_Demyelination/Didactic/MSMicroPath.htm



Normal



Multiple
Sclerosis

Normal resting astrocyte near normally myelinated axons (left)

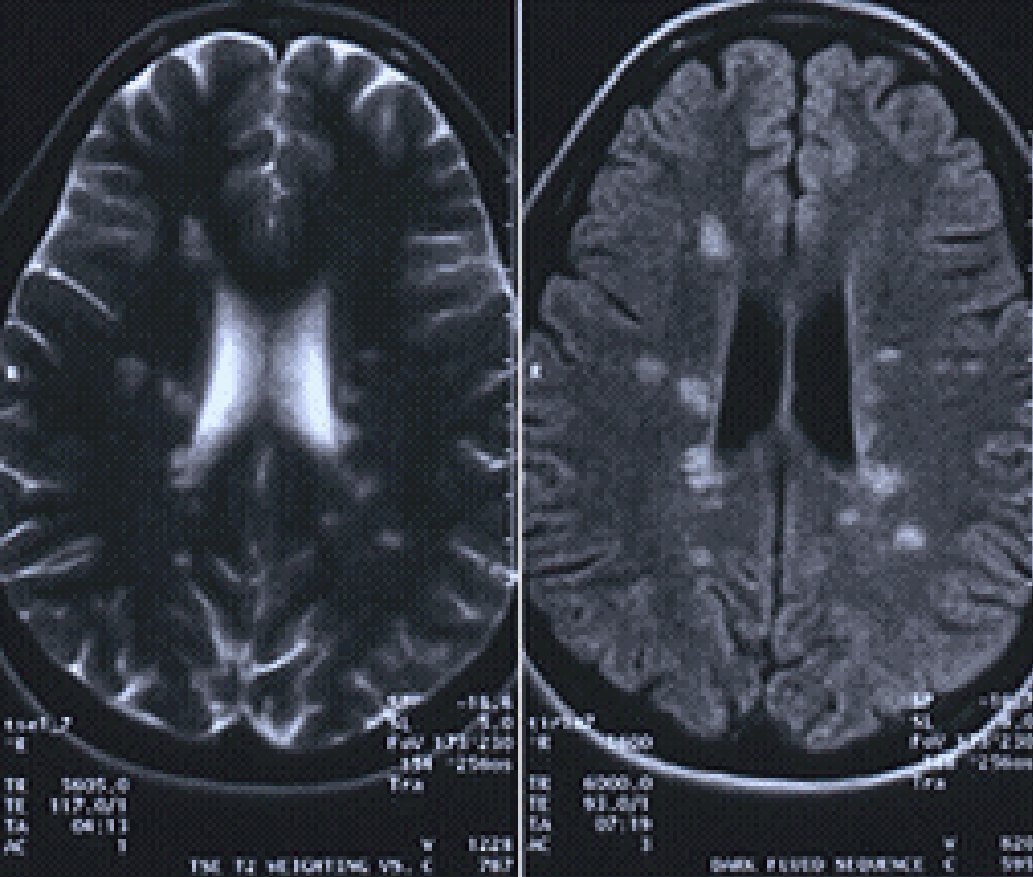
Large reactive astrocytes around demyelinated axons in MS (right)

Multiple Sclerosis

- Distinct episodes of neuro deficits, **separated in time**, attributable to white matter lesions **separated in space**
- Common *presenting symptoms*:
 - weakness in one or more limbs (40%); spinal cord lesions result in motor and sensory impairment of trunk and limbs, spasticity
 - Optic neuritis (22%) – **Frequent initial manifestation**: unilateral visual impairment due to involvement of optic nerve; (10% to 50% of patients, depending on population) go on to develop sensory disturbances (21%)
 - diplopia (12%)
 - vertigo (8%)
 - bladder dysfunction (5%)
 - Brainstem involvement produces cranial nerve signs (ataxia, nystagmus)
- Symptoms may worsen when ambient temperature is higher than usual (**Uhthoff's phenomenon**)
- **Lhermitte's sign** – electrical sensation that runs down back when bending neck

Multiple Sclerosis

- **CSF abnormalities:** mildly elevated protein levels, moderate pleocytosis 1/3 cases (increase mononuclear wbc's), increased IgG levels, and **oligoclonal IgG bands** (immunoelectrophoresis) - presence of a small number of activated B-cell clones in the CNS, thought to be self-reactive
- Evoked response testing may detect slowing in or interruption of conduction in visual, auditory, somatosensory or motor pathways.
- Magnetic resonance imaging – to follow disease progression
- Treatment consists of several types of immunosuppressive or immunomodulatory agents; may slow the progression of the disease but not curative
- Key to diagnosis is recognizing signs and symptoms that would represent multiple lesions, separated by time and in different locations.



MRI is preferred method of imaging the brain to detect plaques or scarring caused by MS. Often, brains that appear to be normal on CT images will show MS plaques on MRI

The diagnosis of MS cannot be made only on the basis of MR imaging. Other neurologic diseases cause lesions in the brain that appear similar.

Normal MR does not absolutely rule out a diagnosis of MS.

Approximately 5% of patients who are confirmed to have multiple sclerosis on the basis of other criteria, do not show any plaques or lesions in the brain on MR images.

Multiple Sclerosis

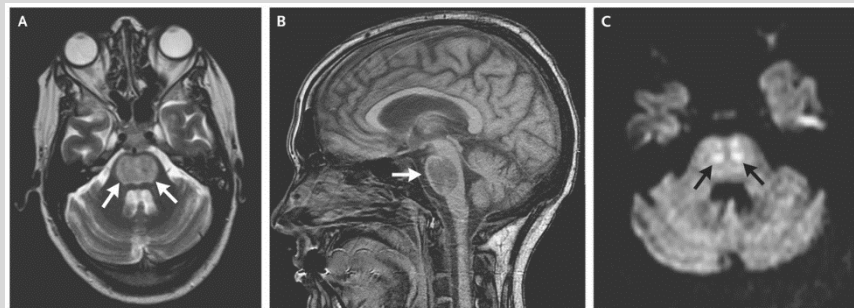
- **Relapsing-Remitting (85%)**
- **Secondary-Progressive**
- **Primary-Progressive**
- **Progressive-Relapsing**
- **Discrete attacks/ flares; resolve over weeks to months without progression**
- **Relapsing-remitting initially, then gradual (chronic) progression resulting in permanent disability**
- **Chronic progression from initial diagnosis**
- **Chronic progression from initial diagnosis but also acute exacerbations – Term not currently used**

Miscellaneous CNS demyelinating diseases

- **Neuromyelitis optica**
 - Syndrome with synchronous (or near-synchronous) bilateral optic neuritis and spinal cord demyelination
 - Once considered a variant of MS, now considered distinct disorder
 - More commonly associated with poor recovery from the first attack
 - Often characterized by the presence of aquaporin-4 antibodies
- **Acute disseminated encephalomyelitis**
 - Diffuse, monophasic demyelinating disease, follows viral infection or, rarely, viral immunization
 - Symptoms usually 1 or 2 weeks later - include headache, lethargy, and coma
 - Rapid clinical course, up to 20% die; remaining patients usually recover completely.
- **Acute necrotizing hemorrhagic encephalomyelitis**
 - Fulminant syndrome of CNS demyelination, usually young adults and children
 - Preceded by recent URI, of unknown cause
 - Fatal in many patients, with significant deficits present in most survivors.

Central pontine myelinolysis (osmotic demyelination syndromes)

- Acute - loss of myelin in base of pons and portions of the pontine tegmentum, roughly symmetric pattern
- Commonly occurs 2 to 6 days after **rapid correction of hyponatremia**; can also be associated with other **severe electrolyte disturbances or osmolar imbalances**
- Rapid increases in osmolality damage oligodendrocytes through uncertain mechanisms
- Inflammation absent from lesions; neurons and axons well preserved
- Because of the synchronous onset of damage, all lesions appear to be at same stage of myelin loss and reaction
- Can involve most parts of brain, periventricular and subpial regions are spared, extremely rare to extend below the pontomedullary junction
- **Clinically rapidly evolving quadriplegia, which may be fatal or lead to severe long-term deficits, including the “locked-in” syndrome - patients are fully conscious yet unresponsive**
- Take home message: **hyponatremia must be corrected slowly and carefully**



Case question

- A 41 year old woman had a flu-like illness for one week. A few days later, she experienced a **rapidly progressive, ascending motor weakness** that required intubation and mechanical ventilation. A CSF analysis shows clear, colorless spinal fluid under normal pressure with a **slightly elevated protein concentration**, normal glucose levels, and a cell count with only a few mononuclear cells. The patient recovered in 3 weeks. If **lymphocytic infiltrates** were seen in peripheral nerves along with **segmental demyelination** at the initial time she saw the physician, what would be the most likely diagnosis?
- A. Multiple sclerosis
- B. Amyotrophic lateral sclerosis
- C. Varicella-zoster virus infection
- D. Guillain-Barre syndrome
- E. Vitamin B 12 deficiency

Guillain-Barré Syndrome (Acute Inflammatory Demyelinating Polyneuropathy)

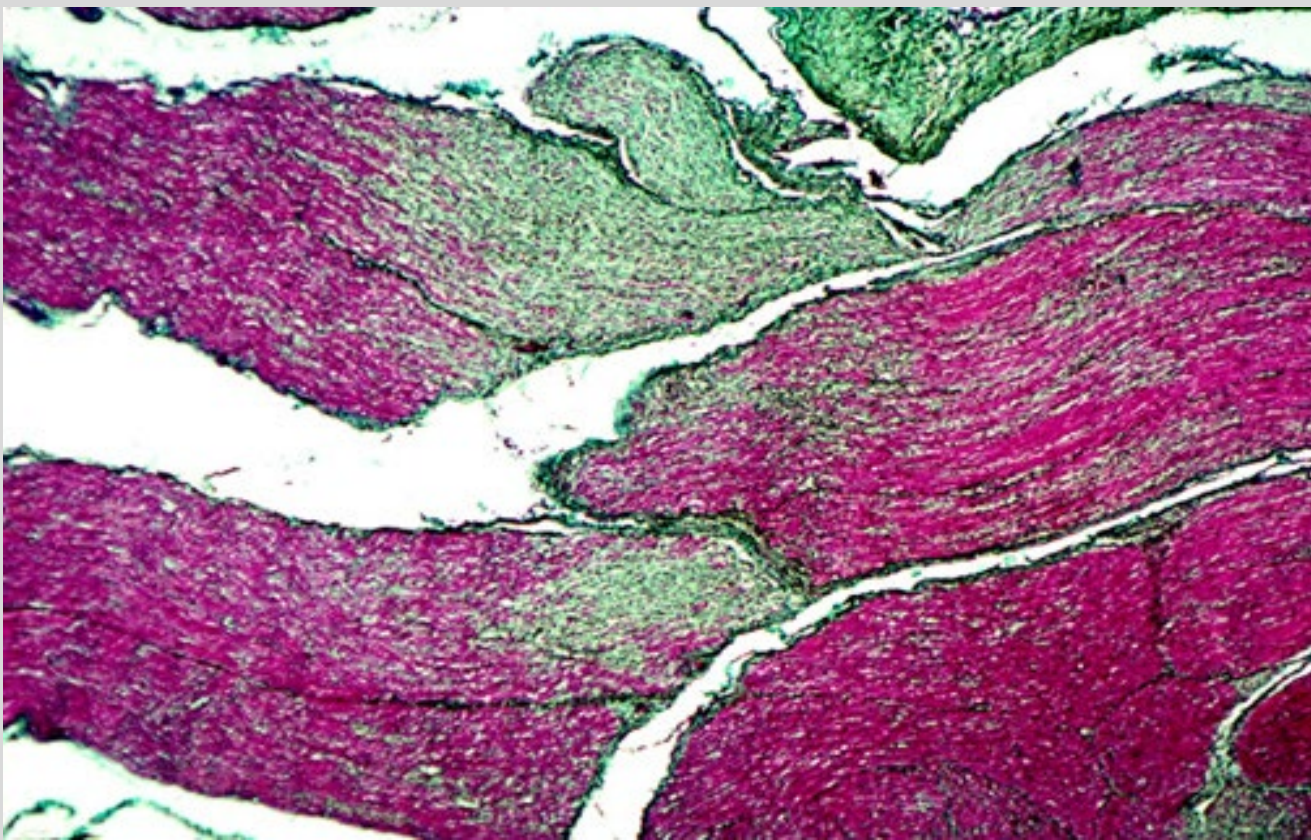
- Acute-onset, immune mediated demyelinating peripheral neuropathy
- May lead to life-threatening respiratory paralysis
- Clinical - weakness beginning in the distal limbs, rapidly advances to proximal muscles (**“ascending paralysis”**)
- Histology - **inflammation and demyelination of spinal nerve roots and peripheral nerves**
- Two-thirds of cases preceded by an acute, influenza-like illness
- Infections with *Campylobacter jejuni*, cytomegalovirus, Epstein-Barr virus, and *Mycoplasma pneumoniae* and also prior vaccination significantly associated with GBS
- No infectious agent has been demonstrated in affected nerves

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 27

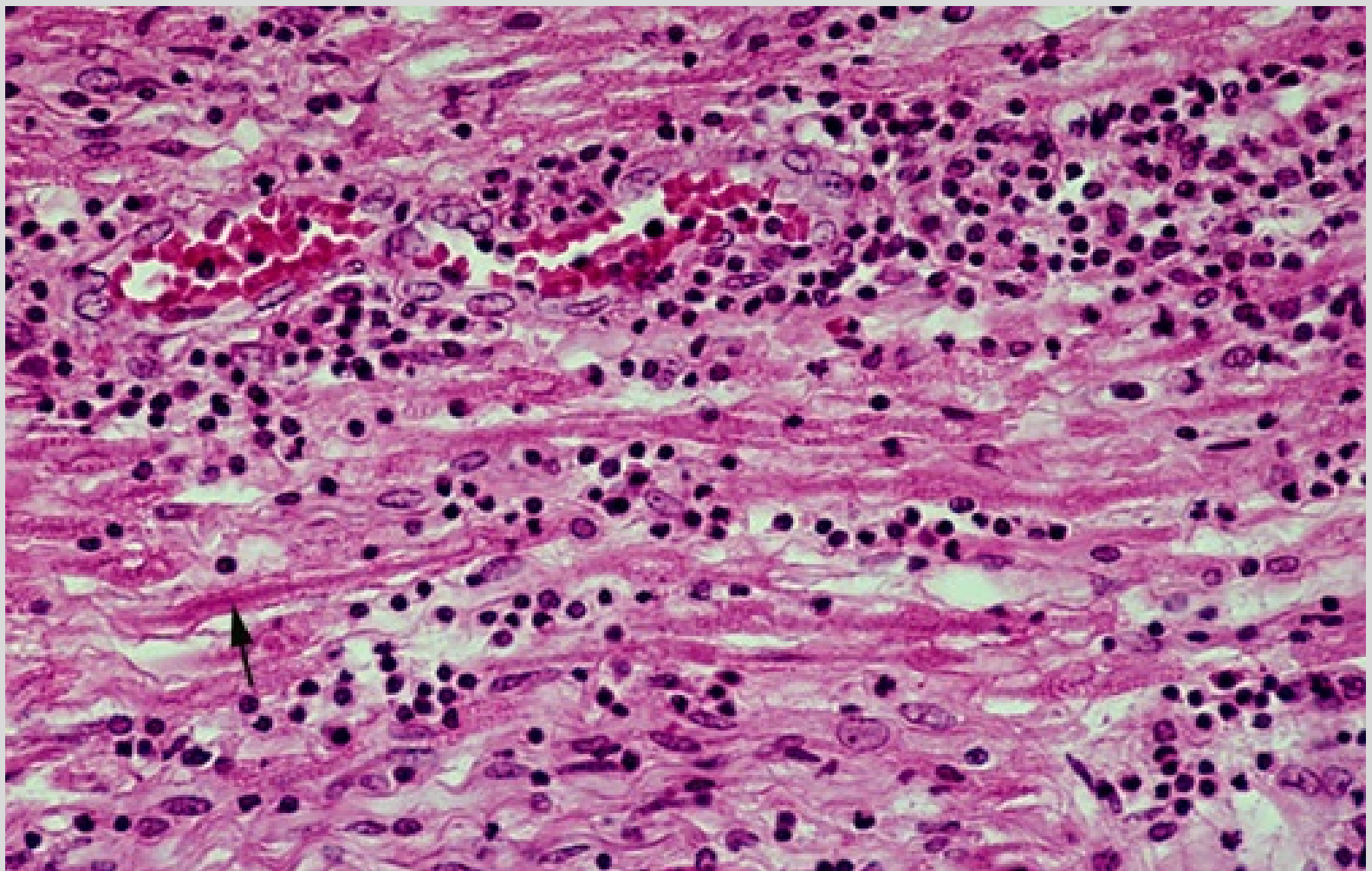
GBS

- Inflammation of peripheral nerves, perivenular and endoneurial infiltration by lymphocytes, macrophages, and few plasma cells.
- **Segmental demyelination** affecting peripheral nerves most prominent lesion; if severe - damage to axons but mostly preserved, Schwann cell preservation
- EM shows early effect on myelin sheaths: cytoplasmic processes of macrophages penetrate basement membrane of Schwann cells, especially near nodes of Ranvier, and extend between the myelin lamellae, stripping the myelin sheath from the axon.
- Ultimately, the remnants of the myelin sheath are engulfed by the macrophages. **Inflammation and demyelination** can be widespread in the peripheral nervous system but are most prominent proximally, close to the nerve roots.

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch. 27



Peripheral nerve in GBS, stained with a myelin stain (pink). Large areas of myelin loss in the center.



- Nerve H&E. Infiltrating inflammation mostly lymphocytes; some residual myelinated axons seen. Denser pink lines (black arrow) are axons and bubbly-appearing pink areas surrounding them are myelin sheath

Guillain-Barre Syndrome- Clinical

- Ascending paralysis and areflexia
- Deep tendon reflexes disappear early
- Sensory involvement, (loss of pain sensation), often present but prominent
- Nerve conduction velocities slowed (multifocal destruction of myelin segments in many axons within a nerve)
- Cerebrospinal fluid (CSF) protein levels elevated due to inflammation and altered permeability of microcirculation within spinal roots as they cross subarachnoid space
- Inflammatory cells remain confined to roots; results in little or no CSF pleocytosis

GBS - Clinical

- Many patients spend weeks in the intensive care unit (ICU) before recovering normal function
- With improved supportive respiratory care, cardiovascular monitoring, and prophylaxis against deep venous thrombosis, the mortality rate has fallen
- Plasmapheresis and intravenous immunoglobulin therapy helpful; they remove pathogenic antibodies and suppress immune function
- 2% to 5% of patients die of respiratory paralysis, autonomic instability, cardiac arrest, or related complications
- Up to 20% of hospitalized survivors suffer long-term disability

Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Ch. 27

Leukodystrophies (genetic metabolic disorders)

“Dysmyelinating” diseases; abnormality in formation or stability of myelin, rather than process in which normal myelin is damaged.

Characteristics of leukodystrophies:

Insidious and progressive loss of cerebral function

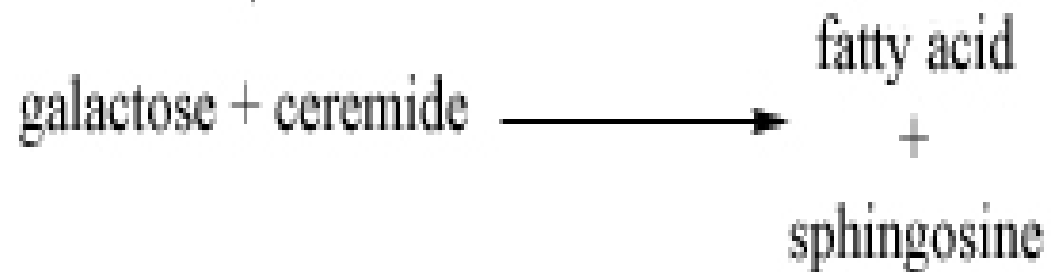
- Age: infancy through adolescence
- **Lysosomal or peroxisomal enzyme defects**
- Involve both CNS and the PNS
- Progressive, resulting in death at early age
- Generally, the earlier the onset of the disease the more severe clinically.

Metachromatic
Leukodystrophy



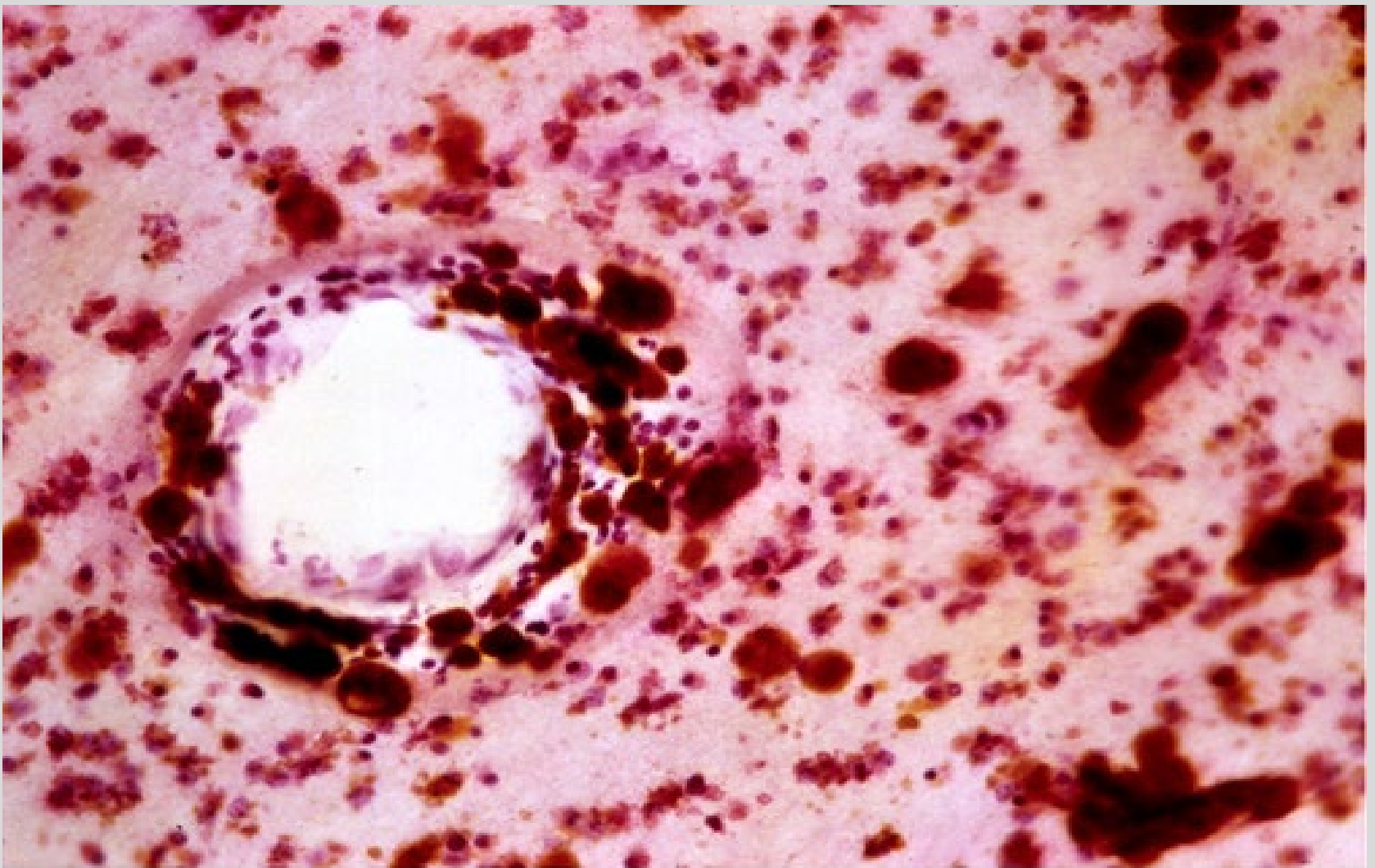
Krabbe's Disease
(Globoid Cell Leukodystrophy)

$\downarrow$
 $//$
galactocerebroside
 β -galactocerebrosidase



Metachromatic Leukodystrophy

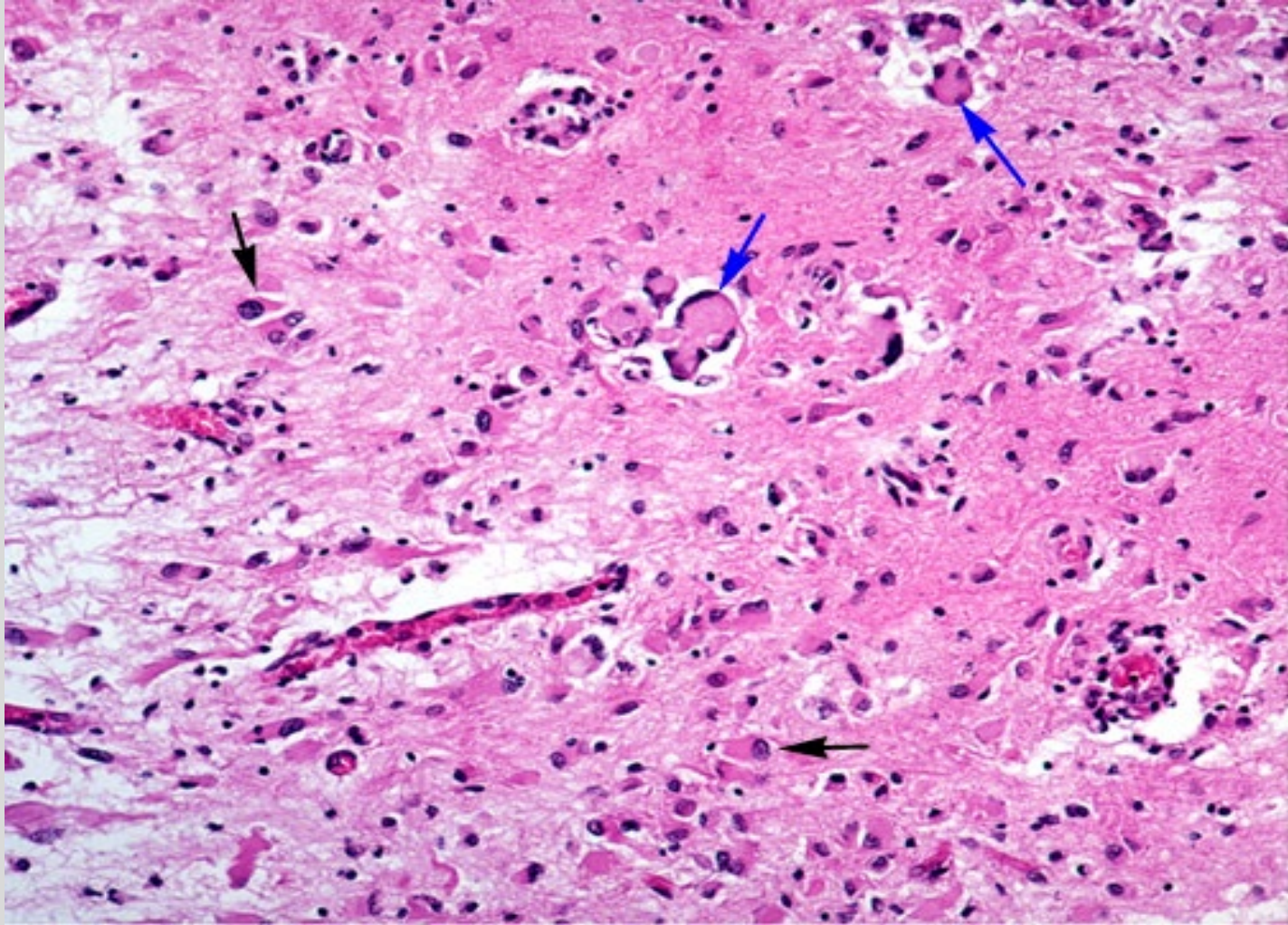
- Autosomal recessive disease - deficiency of lysosomal enzyme, **arylsulfatase A**, which cleaves sulfate from sulfate-containing lipids (sulfatides) as first step in their degradation
- Leads to accumulation of sulfatides, especially cerebroside sulfate, toxic to white matter.
- Histologically - **demyelination with resulting gliosis**
- Macrophages with vacuolated cytoplasm are scattered throughout white matter
- Membrane-bound vacuoles contain complex crystalloid structures made of sulfatides; when they bind to certain dyes (e.g., toluidine blue) sulfatides shift absorbance spectrum of dye, known as **metachromasia**
- **Similar metachromatic material detected in peripheral nerves and in urine – urine test sensitive method for diagnosis.**
- Multiple clinical subtypes



Section of brain with metachromatic leukodystrophy stained with cresyl violet. Sulfatides within the macrophages have bound to the stain.

Krabbe Disease (Globoid Cell Leukodystrophy)

- Autosomal recessive leukodystrophy - deficiency of **galactocerebroside β -galactosidase**, enzyme needed for catabolism of galactocerebroside to ceramide and galactose
- Because of impaired catabolism, alternative catabolic pathway shunts galactocerebroside to galactosylsphingosine
- Elevated levels of galactosylsphingosine toxic to oligodendrocytes and astrocytes
- Brain shows loss of myelin and oligodendrocytes and also affects peripheral nerves (loss of Schwann cells)
- **Diagnostic feature – aggregation of engorged macrophages (globoid cells) in brain and around blood vessels**



Section of a brain with Krabbe's disease: severe astrogliosis (black arrows) and globoid cells around blood vessels (blue arrows).

Adrenoleukodystrophy

- **X-linked recessive disease:** loss-of-function mutations in ABCD1 (encodes an ATP-binding cassette transporter protein required for transport of fatty acids into peroxisomes)
- Boys present with behavioral changes and adrenal insufficiency
- Loss of ABCD1 leads to inability to catabolize very-long-chain fatty acids within peroxisomes, resulting in **elevated serum levels of VLCFAs**
- Symptoms due to progressive CNS, peripheral nerve, and adrenal disease
- **CNS myelin loss, gliosis and extensive lymphocytic infiltration**
- **Adrenal cortical atrophy** results in hypocortisolism.



Cerebral cortex from ALD patient (myelin stain). Pale white matter (WM) normally would be darkly staining.

Other metabolic genetic diseases

- **Other leukodystrophies** include some with defects in lipid metabolism - have other mechanisms of white matter injury including mutations in genes encoding:
 - Proteins required for myelin formation (Pelizaeus-Merzbacher disease)
 - Intermediate filament protein GFAP (Alexander disease)
 - Translation initiation factor eIF2B (vanishing white matter leukoencephalopathy) .
- **Neuronal Storage Diseases** – large class of diseases with genetic heterogeneity usually caused by enzyme deficits or defects in protein or lipid trafficking within neurons -
 - result in accumulation of storage material within neurons, followed by neuronal death
 - leads to seizures and generalized loss of neurologic function
- **Mitochondrial Encephalomyopathies** - Disorders of energy generation causing a range of neurologic disease, often associated with abnormalities in other tissues. May include:
 - Mitochondrial encephalomyopathy, lactic acidosis, and strokelike episodes (**MELAS**) - recurrent episodes of acute neurologic dysfunction, cognitive changes, muscle weakness, and lactic acidosis
 - Myoclonic epilepsy with ragged red fibers (MERRF) resulting in myoclonus (a seizure disorder) and a myopathy showing ragged red fibers on muscle biopsy
 - Leigh syndrome – Infants have lactic acidemia, arrest of psychomotor development, feeding problems, seizures, extraocular palsies, and weakness with hypotonia; death usually occurs within 1 to 2 years

Case Question

- A 6 month old girl with metachromatic leukodystrophy fails to thrive and dies of respiratory insufficiency. Histological examination of the child's brain at autopsy shows marked accumulation of metachromatic material in the white matter and prominent astrogliosis. Lab studies confirm an accumulation of **cerbroside**. This patient most likely suffered from an inborn error of metabolism, characterized by a deficiency in which of the following enzymes?
- A. galactosidase
- B. mannosidase
- C. N-acetyl-hexosaminidase
- D. arylsulfatase A
- E. phenylalanine hydroxylase

Case question

- An 8 month old boy exhibits severe motor, sensory, and cognitive impairments. Brain biopsy shows a disease of white matter characterized by the accumulation of “globoid cells”. Biochemical studies reveal an absence of galactocerebroside beta-galactosidase activity. What is the appropriate diagnosis?
- A. Alexander disease
- B. Hurler disease
- C. Krabbe disease
- D. Metachromatic leukodystrophy
- E. Adrenoleukodystrophy

Case question

- A 27 year old man presents with dementia, weakness, visual loss, and ataxia one year after receiving a **cadaveric kidney transplant**. Which of the following is the most likely cause of this patient's CNS disorder?
- A. Adrenoleukodystrophy
- B. Gaucher disease
- C. Metachromatic leukodystrophy
- D. Progressive multifocal leukoencephalopathy
- E. Subacute sclerosing panencephalitis

PROGRESSIVE MULTIFOCAL LEUKOENCEPHALOPATHY

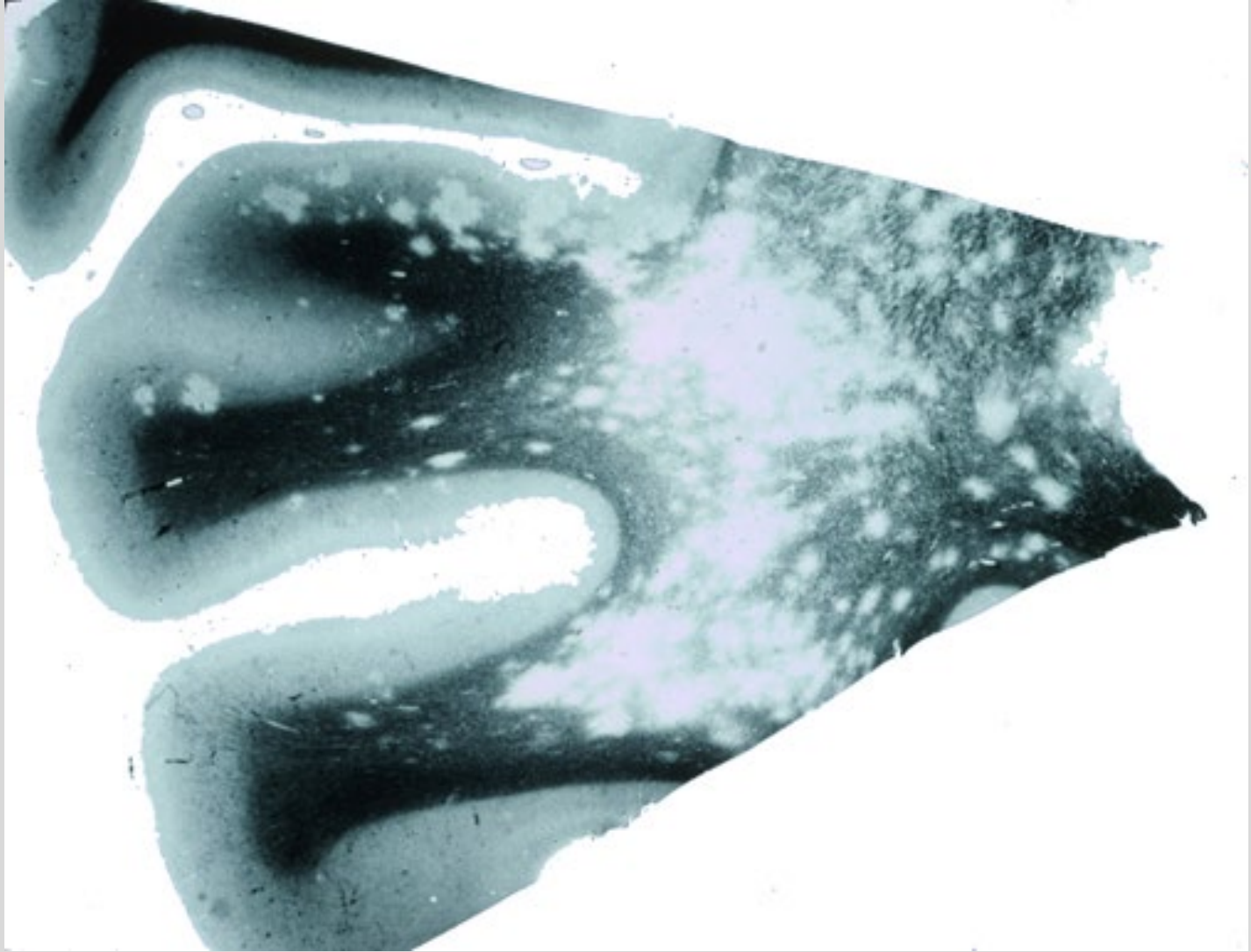
- Viral encephalitis caused by **JC polyomavirus**
- **Virus preferentially infects oligodendrocytes**, therefore demyelination
- Approximately 65% of normal population have serological evidence of exposure to virus by age 14 years. Therefore, virus may be reactivated by immunosuppression

Progressive Multifocal Leukoencephalopathy

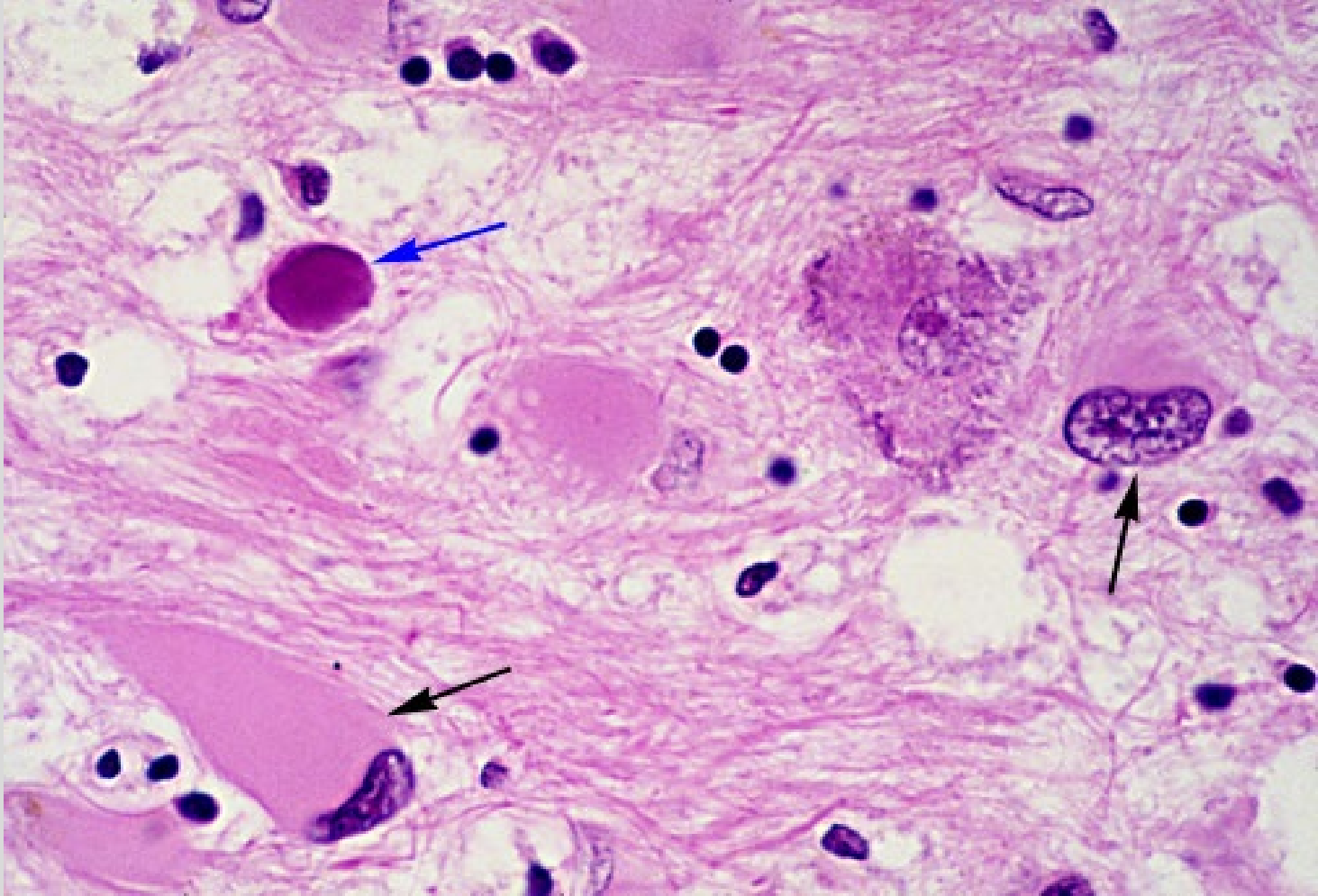
- **Immunosuppressed patients**, including transplant patients and AIDS patients, are at risk. Immunocompetent people usually do not get PML
- Clinical – focal (depending on location), progressive neurologic symptoms and signs
- Extensive, usually multifocal, cerebral or cerebellar demyelinative lesions can be seen on CT and MRI

PML

- Patches of irregular, ill-defined destruction of white matter
- Micro:
 - patches of demyelination, often subcortical
 - In center: usually lipid laden macrophages and decreased number of axons
 - Edge of lesion - very **enlarged oligodendrocyte nuclei**; containing **glassy amphophilic viral inclusion**
 - Astrogliosis, include atypical forms (bizarre, giant astrocytes)

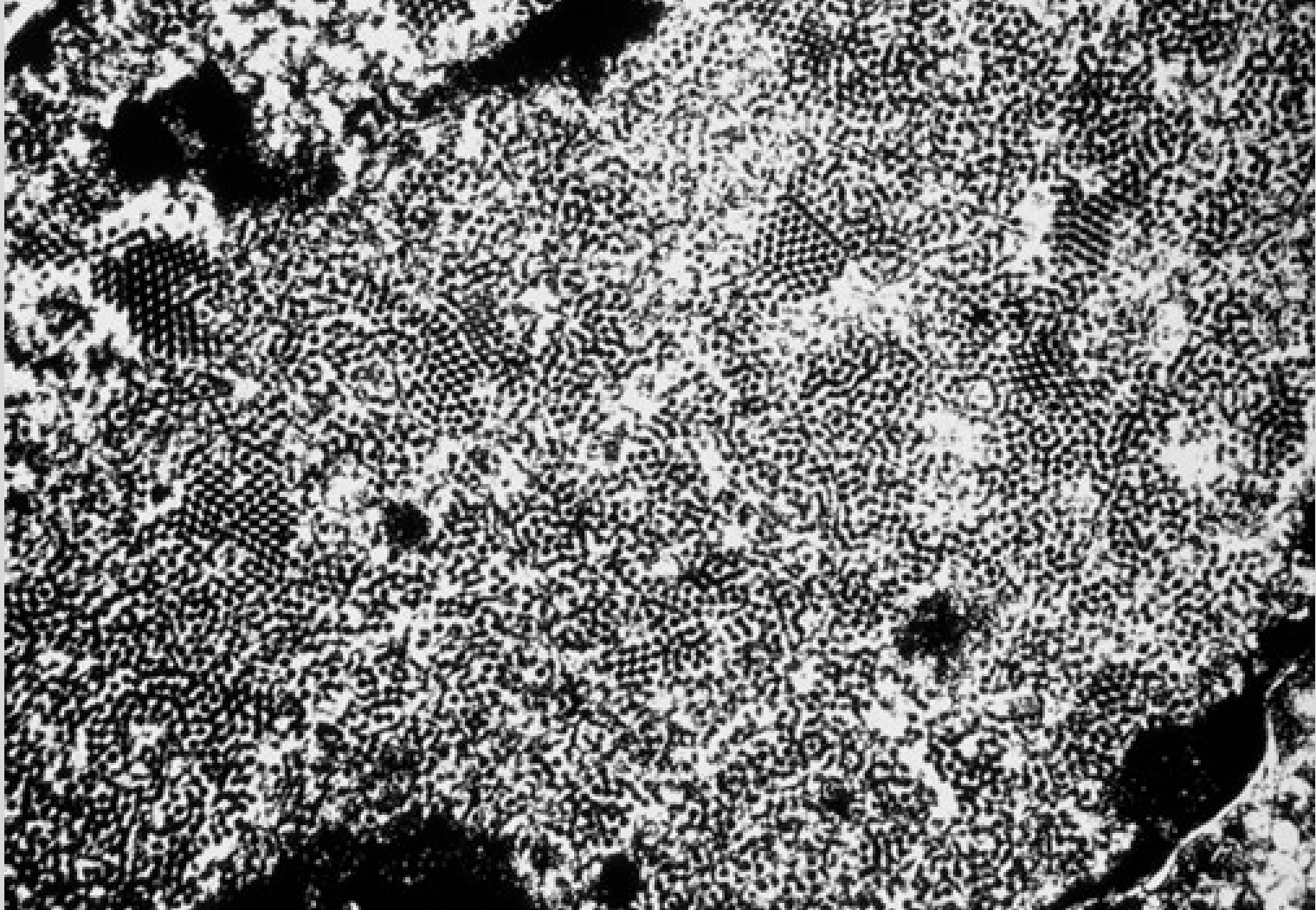


- Cerebral cortex stained for myelin: multifocal demyelination scattered throughout subcortical white matter.



PML (H&E): reactive astrocytes (black arrows), frequently pleomorphic .

Blue arrow = oligodendrocyte nucleus markedly enlarged with viral particles, giving it a magenta color.



EM shows **JC virus particles** in an oligodendroglial nucleus.

Summary Slide

- Review of basic demyelination
- Multiple Sclerosis – chronic, autoimmune, demyelinating disease
 - Environmental and genetic
 - Plaques separated in space and time, inflammation and demyelination
 - Both motor and sensory disturbances
 - CSF – oligoclonal bands
- Guillain-Barre syndrome – acute, immune, inflammation and segmental demyelination, may be preceding viral illness, ascending paralysis

Summary Slide (continued)

- Leukodystrophies
 - Metachromatic leukodystrophy – deficiency in arylsulfatase-A results in sulfatide accumulation which causes myelin breakdown, AR, sulfatides bind stain and changes color
 - Krabbe's disease – deficiency in galactocerebroside beta-galactosidase which shunts to production of galactosylsphingosine, toxic to oligodendrocytes – “globoid cells” around blood vessels
 - Adrenoleukodystrophy – X- linked, lack of catabolism of VLCFAs, atrophy of adrenal gland, behavioural changes
- PML – JC polyomavirus infects oligodendrocytes which causes demyelination, immunocompromised patients

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>